Statistics 652: Fall 2026 *Note: Revised Version.*
Problem Set 2

2.1. Recall the mean can be expressed as the functional $T(F) = E_F(X)$ and for the median can be expressed as the functional $T(F) = F^{-1}(0.5)$.

i) For continuous F, express the trimmed mean and the Windsorized mean in the form of a functional $T(F)$.

ii) For absolutely continuous F, derive the influence function and gross error sensitivity of the median, and evaluate $E[IF(X;T,F)]$ and $E[IF(X;T,F)^2]$.

2.2.a. Show that the mode is *not* qualitatively robust.

2.2.b. Show that the trimmed mean and the Windsorized mean can be expressed as L-statistics. That is, find the corresponding values of $a_1, \ldots, a_n$ in the expression $T_n = \sum_{i=1}^n a_i x_{(i)}$.

2.3. For a location M-estimate, show: (i) $\rho(x,t) = \rho(x-t)$ with $\rho(-r) = \rho(r)$, and
(ii) $\psi(x,t) = \psi(x-t)$ with $\psi(-r) = -\psi(r)$.

2.4. Derive the asymptotic relative efficiency of the mean to the median at

i) the location scale t-family, i.e. when $X \sim \mu + \sigma t_\nu$ where $\mu \in \mathrm{R}, \sigma > 0$ and t_ν is the standard t-distribution on ν degrees of freedom. (That is, verify the formula on page 6 of the robust lecture notes.)

ii) At the mixture of normal distribution $X \sim (1-\epsilon)Normal(\mu, \sigma^2) + \epsilon Normal(\mu, 9\sigma^2)$.

2.5. Consider a one parameter M-estimator $\widehat{\theta}_n$ defined as the minimizer of $\sum_{i=1}^n \rho(x_i;\theta)$. When $\rho(x;\theta)$ corresponds to the negative log-likelihood for a regular parametric family of distributions $F_\theta = F(x;\theta)$, show that $E_F[IF(x;T,F)^2] = 1/I(\theta;X)$ when evaluated at $F = F_\theta$. *Show this directly. Do not use the asymptotic theory of maximum likelihood estimators.*

2.6. Let F denote the distribution function of $X \in \mathrm{R}$. Presume the second moment of F exists, and let σ_F^2 denote the standard deviation of the distribution F.

i) Express σ_F^2 as a functional, i.e. as $T(F)$.

ii) Find the influence function of σ_F^2, i.e. find $IF(x;T,F)$. Also, find its gross error sensitivity and sketch the influence function as a function of $x \in \mathrm{R}$.

iii) For this case: (a) verify $E_F[IF(X;T,F)] = 0$, and (b) find $E_F[IF(X;T,F)^2]$.

iv) Let $X_1, \ldots, X_n$ be a random sample from F. Presume the fourth moment of F exists, and consider the sample variance $s_n^2 = n^{-1}\sum_{i=1}^n (X_i - \overline{X})^2$. The asymptotic variance of s_n^2 is defined as the variance of the limiting normal distribution of $\sqrt{n}(s_n^2 - \sigma)$. Use the CLT to derive the asymptotic variance of s_n^2, and verify whether of not it is equal to $E_F[IF(X;T,F)^2]$.

2.7. Let F denote the distribution function of $X \in \mathrm{R}$. Presume the second moment of F exists, and let σ denote the standard deviation of the distribution F.

i) Express σ as a functional, i.e. as $T(F)$.

ii) Find the influence function of σ, i.e. find $IF(x; T, F)$. Also, sketch the influence function as a function of $x \in \mathrm{R}$.

iii) Let $X_1, \ldots, X_n$ be a random sample from F. Presume the fourth moment of F exists, and consider the sample standard deviation $s_n = \sqrt{n^{-1} \sum_{i=1}^{n} (X_i - \overline{X})^2}$. The asymptotic variance of s_n is defined as the variance of the limiting normal distribution of $\sqrt{n}(s_n - \sigma)$. Verify whether of not this asymptotic variance is equal to $E_F[IF(X; T, F)^2]$.

2.8. Derive the density of a scale mixture of normals. This is, for $\mu \in \mathrm{R}$ and $\sigma > 0$ derive the density of $X = \mu + \sigma RZ$, where Z has a standard normal distribution, R is a positive random variable with density $f(r)$, and with R and Z are independent.

2.9. Optional: Prove the consistency and asymptotic normality of the sample median. That is, if $\widehat{\mu}_n$ and μ represent the sample and population median respectively, and the underlying distribution has density $f(s)$ with $f(\mu) > 0$, then show that $\sqrt{n}(\widehat{\mu}_n - \mu) \to_d N(\mu, 1/\{2f(\mu)\}^2)$.

2.10. Show that if X_1 and X_2 have independent $Cauchy(\mu, \sigma^2)$ distributions, then $\overline{X} = (X_1 + X_2)/2$ has a $Cauchy(\mu, \sigma^2)$ distribution.

2.11. A random variable X is said to have a symmetric stable distribution if when X_1 and X_2 are independent copies of X, then $X_1 + X_2 \sim_d aX$ for some constant a. A standardized symmetric stable distribution with index $0 < \alpha \leq 2$ has characteristic function $\phi(t) = E(exp\{itX\}) = exp\{-|t|^\alpha\}$. If $X_1, \ldots, X_n$ represent a random sample from such a distribution, and $Y_i = \mu + \sigma X_i$ for $i = 1, \ldots, n$, find the characteristic function of $\overline{Y} = \frac{1}{n} \sum_{i=1}^{n} Y_i$. Comment on how the spread of $\overline{Y}$ depends on the sample size n.

2.12. Prove that $E_F(IF(X; T, F)) = 0$ whenever T is an M-functional. (Note: proving this equality for a general functional T is mathematically advanced.)

2.13. Show that Tukey's M-estimate of location $\widehat{\theta}$ does not correspond to the maximum likelihood estimate of location $\theta \in \mathrm{R}$ under any symmetric symmetric location model, i.e. one for which the density satisfies $f(x - \theta) = f(\theta - x)$.

2.14. Show that the MLE for the location-scale double exponential distribution show that the scale estimate does not have a bounded influence function. Also, find the finite sample breakdown point for the scale estimate.

2.15. Find the MLE for $\boldsymbol{\beta}$ and σ^2 for the regression model $Y_i = \mathbf{X}_i' \boldsymbol{\beta} + \epsilon_i$ where ϵ_i, $i = 1, \ldots, n$ are i.i.d. $N(0, \sigma^2)$ using the theory of exponential families.

2.16. Verify the adaptively weighted least squares representation for an M-estimate of $\widehat{\boldsymbol{\beta}}$ given on page 50 of the robustness slides.